ModelIt!

Teacher's Guide

How Does Water Shape the Land?

The Power of Weathering and Erosion

Quick Reference

Grade Level	Grade 3
NGSS Standard	3-ESS2-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 3-ESS2-1

Represent data in tables and graphical displays to describe typical weather conditions expected during a particular season.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model showing how water flow, erosion strength, and land shape change are connected, then use simulations to test how different water flows affect erosion.
Disciplinary Core Idea	**ESS2.A Earth Materials and Systems** — Wind and water can change the shape of the land. Weathering and erosion by water, ice, wind, and vegetation break rocks, soils, and sediments into smaller particles and move them around.
Crosscutting Concept	**Cause and Effect** — Students investigate how increasing water flow (cause) leads to stronger erosion and greater land shape changes (effects) in a clear cause-and-effect relationship.

CAST Testing Alignment

MULTIPLE CHOICE:

After a heavy rainstorm, a student notices that a path of mud and small rocks has formed going downhill from the school playground. Which process BEST explains what happened?

CONSTRUCTED RESPONSE:

Explain how a river can carve a canyon over time. Use the words 'water flow,' 'erosion,' and 'land shape' in your answer to describe the process step by step.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Earth's surface is constantly changing, shaped by powerful forces that work day and night, year after year. Weathering and erosion are two of the most important processes that sculpt our planet's landscapes.

Weathering: Breaking Rocks Apart

Weathering is the process of breaking rocks into smaller and smaller pieces. Physical weathering happens when water seeps into cracks, freezes, expands, and splits the rock (called frost wedging). Plant roots grow into cracks and push rocks apart. Chemical weathering happens when rainwater (which is slightly acidic) slowly dissolves certain minerals in rocks. Over time, even the hardest granite mountain is broken down into sand and soil by these relentless processes.

Erosion: Moving Material Around

Erosion is the transportation of weathered material from one place to another. Water is the most powerful erosion agent — rivers carry billions of tons of sediment to the ocean every year. Wind picks up fine particles and blows them across deserts. Glaciers (giant rivers of ice) grind valleys as they slowly flow downhill. Gravity pulls loose rocks downslope in landslides. The faster the agent moves, the more material it can carry.

Deposition: Building New Landforms

When the erosion agent slows down, it drops the material it was carrying — this is deposition. Rivers deposit sediment at their mouths, forming deltas. Wind deposits sand in dunes. Ocean waves deposit sand on beaches. The Mississippi River delta was built entirely from sediment the river carried from as far away as Montana. Erosion takes material from one place and deposition builds it up somewhere else — constantly reshaping Earth.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System

Students identify water flow as an external component and erosion strength and land shape change as internal components that respond to how much water is moving across the land.

E — Establish Relationships

Students determine that water flow positively affects erosion strength, and erosion strength positively affects land shape change — more water means more erosion means bigger changes to the land.

V — Visualize & Model

Students build a model showing how increasing water flow leads to stronger erosion and more dramatic changes in land shape over time.

E — Evaluate Outcomes

Students run gentle stream, steady rain, and flooding river scenarios to observe how changing water flow affects erosion and land shape across the system.

R — Revise & Extend

Students add slope steepness or vegetation cover to explore how hills and plants affect the erosion process — steeper slopes erode faster, plants slow erosion down.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Look at this canyon — it is a MILE deep! How do you think it got there? A giant shovel? An earthquake? Nope — just water!"*

Watch For

Show photos of the Grand Canyon and other water-carved landscapes. Ask students to guess how they formed.

Slide 2: What Will We Learn Today?

3 min

Say: *"Today we are going to discover how something as simple as water can carve through solid rock and reshape entire mountains!"*

Watch For

Read objectives together. Introduce vocabulary — pour a small amount of water on a pile of sand to show erosion in action.

Slide 3: The Big Question

3 min

Say: *"Water is soft. Rock is hard. So how can water possibly carve through rock? This is our mystery!"*

Watch For

Have students feel water and a rock. Ask: Which is harder? Which is softer? So how did WATER carve the Grand Canyon?

Slide 4: Let's Build a Model!

3 min

Say: *"We are going to build an erosion model to see how water really does change the land!"*

Watch For

Preview each LEVER step. Show a quick before-and-after of a landscape that changed due to erosion.

Slide 5: Activity 1: Sort the Parts

7 min

Say: *"What parts of this erosion system can we identify? What drives the whole process?"*

Watch For

Guide sorting of three components. Explain water flow is external because rainfall and rivers bring water to the land.

Slide 6: Activity 2: Connect with Arrows

7 min

Say: *"When water flow gets stronger — like during a flood — does erosion increase or decrease?"*

Watch For

Help students draw positive arrows from water flow to erosion strength, and from erosion strength to land shape change.

Slide 7: Activity 3: Run the Simulation!

8 min

Say: *"Let us test it! We will try a gentle stream, steady rain, and a powerful flood. What do you predict?"*

Watch For

Have students predict first, then run each scenario. Compare the amount of erosion and land change in each.

Slide 8: What Did We Discover?

7 min

Say: *"Water is the most powerful sculptor on Earth! Given enough time, it can carve through anything."*

Watch For

Summarize discoveries. Quick demo: drip water onto a sugar cube to show how water dissolves and erodes material. Connect to the Grand Canyon.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Water Flow
- **INTERNAL:** Erosion Strength, Land Shape Change

Relationships

- **Water Flow to Erosion Strength:** POSITIVE (+) — Faster, stronger water flow picks up more rock and soil. A gentle stream might carry tiny sand particles, but a flooding river can move boulders. The more water flowing, the more powerful the erosion force.
- **Erosion Strength to Land Shape Change:** POSITIVE (+) — Stronger erosion removes more material, creating bigger changes in the land. Weak erosion might make tiny grooves, but strong erosion over time can carve mile-deep canyons and reshape entire coastlines.

Key Discoveries

- Faster and stronger water flow causes more erosion — floods can move huge boulders and carve deep channels
- Even gentle water flow causes erosion over long periods of time — dripping water carved the Grand Canyon over millions of years
- Erosion picks up material and deposition drops it off — this is how deltas, beaches, and sandbars form
- Weathering and erosion are constantly changing the shape of Earth's surface, even when we cannot see it happening

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Erosion only happens during floods and storms"	Erosion happens constantly, even on sunny days. Rivers, streams, and waves are always carrying sediment. Wind erodes deserts every day. Even dew and frost can cause tiny amounts of weathering. Big storms cause dramatic erosion, but the slow, everyday erosion adds up to enormous changes over time.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Rocks last forever and never change"	Rocks seem permanent, but they are constantly being broken down by weathering. Water, ice, wind, plant roots, and even temperature changes slowly crack, dissolve, and crumble rocks. A granite mountain that looks unchanged today will be worn down to a hill in a few million years. No rock lasts forever.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Water only erodes soft things like sand and dirt"	Water can erode any material given enough time, including the hardest rocks. The Grand Canyon is carved through layers of limestone, sandstone, and even hard granite. Water gets into microscopic cracks, freezes and expands, and slowly breaks the rock apart. Chemical reactions between water and minerals also dissolve rock from the inside out.